exampro

Name:

Class: Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.7 Magnetism and Electromagnetism

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

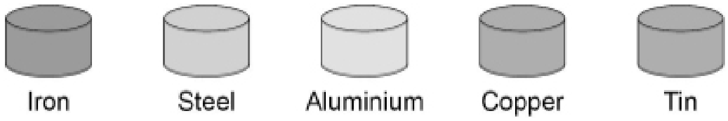
- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

01.1 **Figure 1** shows five different metal samples.

Figure 1



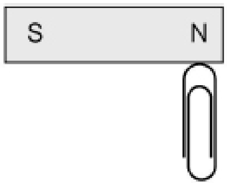
A student placed a magnet close to each metal sample.

Describe what happened.

[2 marks]

01.2 **Figure 2** shows a paper clip being attracted to a permanent magnet.

Figure 2



The paper clip in the diagram is not a permanent magnet.

Explain what would happen if the paper clip was removed and brought close to the south pole of the permanent magnet.

[2 marks]

02.1 **Figure 3** shows three metal blocks.

The blocks are not labelled.

One block is a permanent magnet, one is iron and one is aluminium.

Figure 3

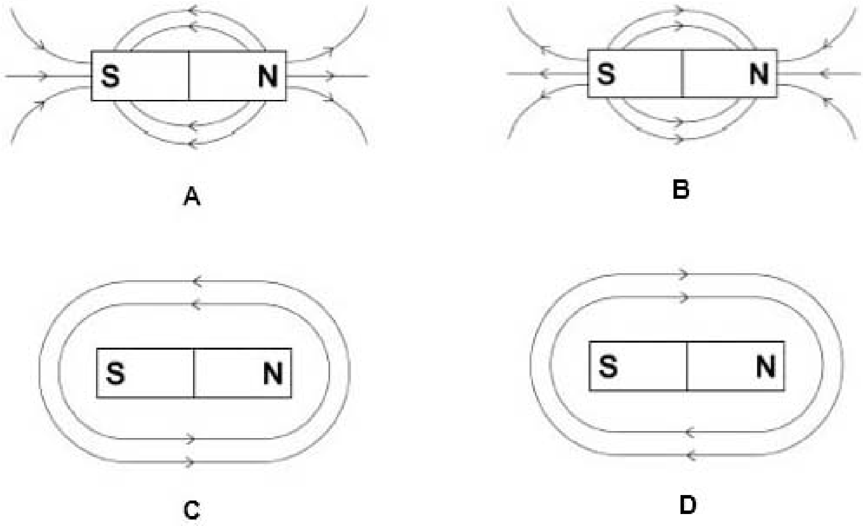
Describe how another permanent magnet can be used to identify the blocks.

[3 marks]

02.2 A magnet produces a magnetic field.

Which diagram shows the magnetic field pattern around a bar magnet?

Tick **one** box.

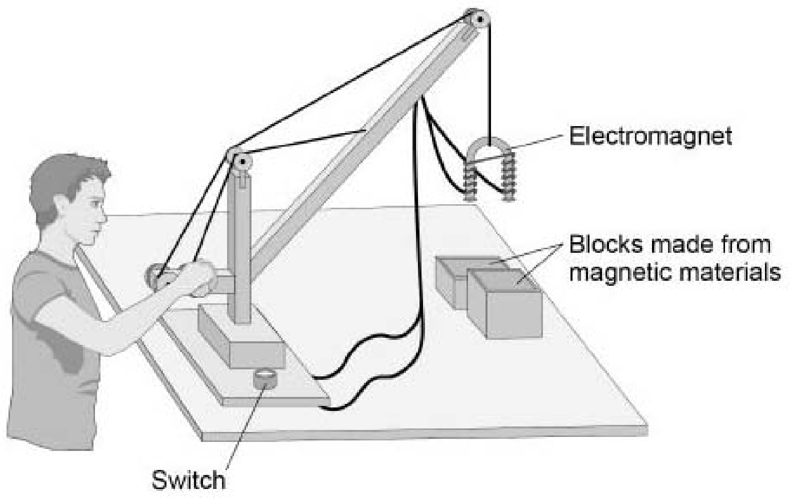


- ☐ A
- ☐ B
- ☐ C
- ☐ D

[1 mark]

02.3 **Figure 4** shows a toy crane.

Figure 4



The toy crane uses an electromagnet to pick up and move the blocks.

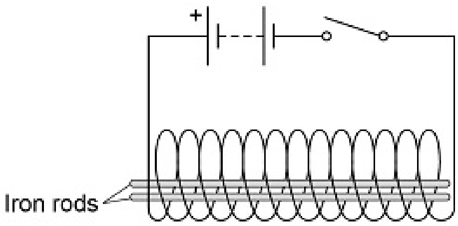
Explain how this electromagnet is able to pick up and move the blocks.

[6 marks]

03

Figure 5 shows two iron rods placed inside a solenoid.

Figure 5



Explain why the iron rods move apart when the switch is closed.

[2 marks]

04

Figure 6 shows a portable power supply.

Figure 6

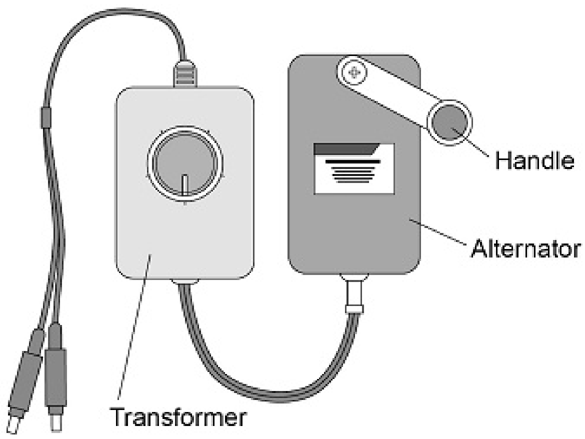
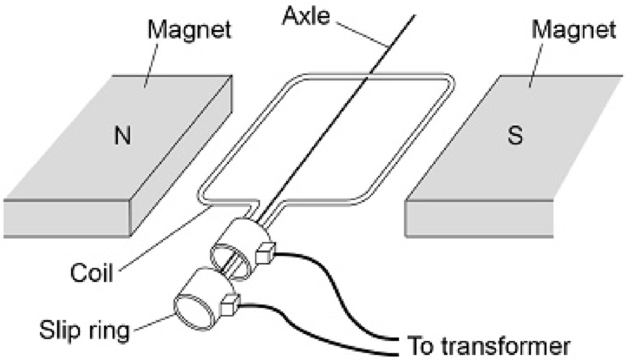


Figure 7 shows the inside parts of the alternator.

Figure 7



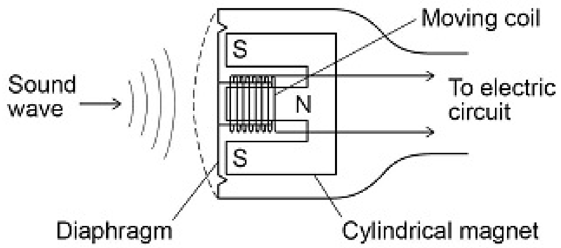
The handle of the alternator is turned, causing the coil to rotate.

Explain why an alternating current is induced in the coil.

[5 marks]

05 **Figure 8** shows the parts of a moving-coil microphone.

Figure 8

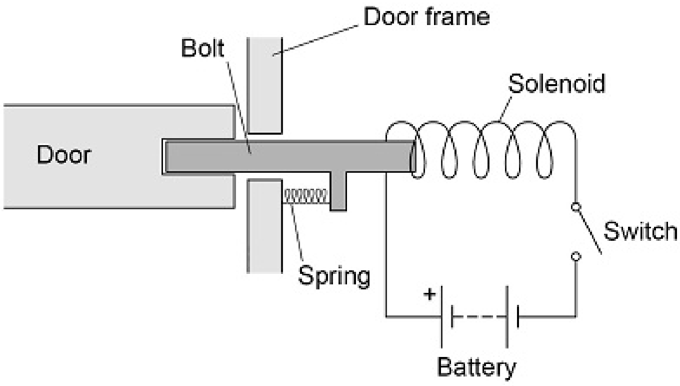


Explain how a moving-coil microphone works.

[4 marks]

06.1 **Figure 9** shows a lock. The door unlocks when the switch is closed.

Figure 9



Which material should the bolt be made from?

Tick **one** box.

☐ Aluminium

☐ Brass

☐ Copper

☐ Iron

[1 mark]

06.2 Explain why the door unlocks when the switch is closed.

[3 marks]

06.3 Give **two** ways the resultant force on the bolt could be increased.

1. ____________________

____

2. ____________________

____

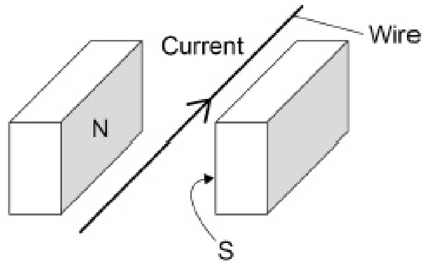
[2 marks]

07

Figure 10 shows a wire in a magnetic field.

The direction of the current in the wire is shown.

Figure 10



There is a force on the wire due to the current in the magnetic field.

In which direction is the force on the wire?

A **B** **C** **D**

Tick **one** box.

☐ A

☐ B

☐ C

☐ D

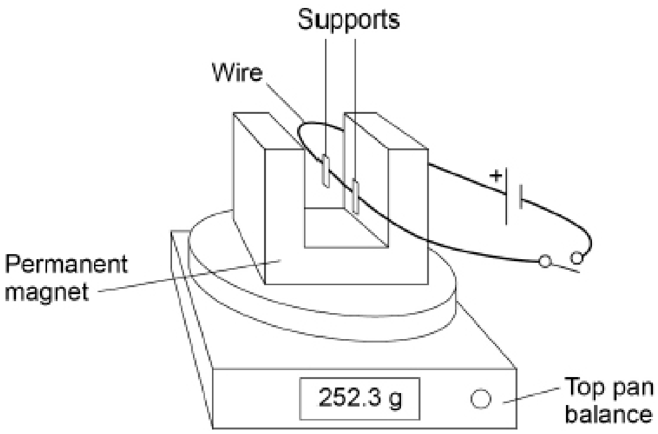
[1 mark]

08.1 A student clamped a wire between the poles of a permanent magnet.

The student investigated how the force on the wire varied with the current in the wire.

Figure 11 shows the equipment used.

Figure 11



The top pan balance was used to determine the force on the wire.

When the switch was closed, the reading on the top pan balance increased.

Explain why the increased reading showed that there was an upward force on the wire.

[2 marks]

08.2 **Table 1** shows the readings on the top pan balance with the switch open and with the switch closed.

Table 1

Switch	**Mass in grams**
Open	252.3
Closed	254.8

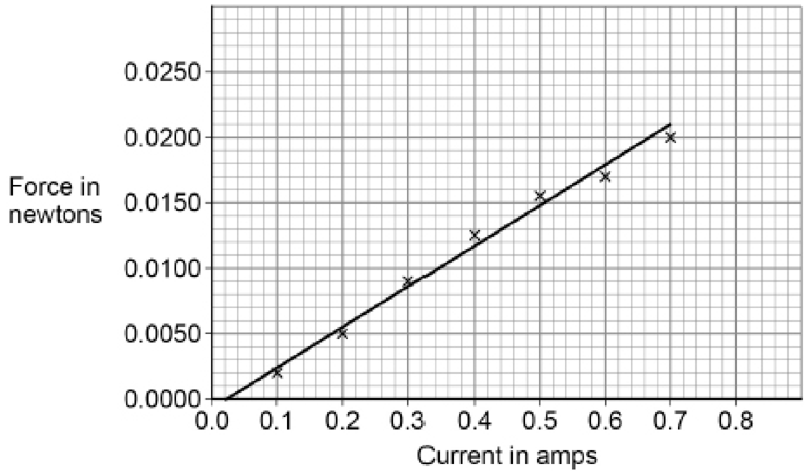
Explain how the values in the table above can be used to determine the size of the force on the wire.

[2 marks]

08.3 The student varied the current in the wire and calculated the force acting on the wire.

Figure 12 shows the results.

Figure 12



The length of the wire in the magnetic field was 0.125 m.

Determine the magnetic flux density.

Magnetic flux density = ____________ T

[4 marks]